

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R,S) / S6 (PT) / (WP) Exam April 2025 (2019 Scheme)

Course Code: EET302**Course Name: LINEAR CONTROL SYSTEMS**

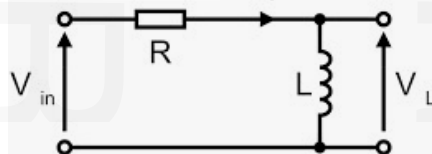
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Define transfer function and find the transfer function of the given RL network. (3)



- 2 Give the characteristics of a lead compensator. (3)
- 3 Draw the unit step response of a first order system. (3)
- 4 A unity feedback system has an open loop transfer function $G(s)H(s) = \frac{s+2}{s(s+1)(s+10)}$. Determine the static error coefficients and steady state errors. (3)
- 5 Explain magnitude and angle criterion referred to root locus? (3)
- 6 Explain the effect of adding poles and zeros on root locus. (3)
- 7 Write the advantages of frequency response analysis. (3)
- 8 Define the phase cross over frequency and gain cross over frequency of a system. (3)
- 9 State the advantages of Nyquist Plot. (3)
- 10 Explain the need for compensation in control system. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Determine the transfer function of a lead-lag compensator using RC circuit components. List the characteristics of the lead-lag compensators. (8)
- b) Explain the working of an AC Servomotor with a schematic diagram. (6)

OR

- 12 a) Derive the transfer function of a field controlled DC motor and represent the system in block diagram form. (8)

- b) Explain the working of a synchro. (6)

Module II

- 13 a) Derive the expression for rise time and settling time of an under damped second order system. (8)
- b) Determine the damping ratio, damped frequency of oscillation, rise time, peak time, peak overshoot and settling time for the system having closed loop transfer function $\frac{C(s)}{R(s)} = \frac{25}{s^2+6s+25}$. (6)

OR

- 14 a) The forward path transfer function of a unity feedback control system is given by $G(s) = \frac{4s+1}{4s^2}$. Determine the unit step response of the system. (8)
- b) Using Routh's stability criterion determine the stability of the given system whose characteristic equation is $s^6+2s^5+8s^4+15s^3+20s^2+16s+16=0$. (6)

Module III

- 15 a) Determine the value of K for which the system with an open loop transfer function $G(s) = \frac{K}{s(s+1)(s+5)}$ exhibit a step response with damping ratio 0.707 using Root locus plot. (10)
- b) Explain PI and PD controllers, with suitable block diagrams. (4)

OR

- 16 a) Consider a unity feedback system with an open loop transfer function, $G(s) = \frac{K}{s(s+2)(s+8)}$. Design a suitable compensator to meet the following specifications. (10)
- (i) Percentage Peak overshoot = 16% for unit step input.
- (ii) Steady state error ≤ 0.125 for unit ramp input.
- b) Explain the PID controllers and its functions in any closed loop control system. (4)

Module IV

- 17 a) Sketch Bode Plot for the system with an open loop transfer function $G(s)H(s) = \frac{1}{s(1+0.02s)(1+0.04s)}$. From Bode plot, determine gain margin and phase margin and assess the stability of the system. (10)
- b) Define resonant frequency, resonant peak, bandwidth and cut off rate for a standard second order system. (4)

OR

- 18 a) Sketch Polar Plot for the given open loop transfer function $G(s)H(s) = \frac{10(s+2)}{s(s+1)(s+3)}$ and determine the gain margin and phase margin (10)

b) Define gain margin and phase margin. (4)

Module V

- 19 Investigate the closed loop stability for the system having an open loop transfer function $G(s)H(s) = \frac{1+4s}{s^2(1+s)(1+2s)}$, using Nyquist stability criterion. (14)

OR

- 20 A unity feedback system has an open loop transfer function $G(s) = \frac{K}{s(1+2s)}$ (14)

Design a suitable lag compensator so that phase margin is 40° and the steady state error for ramp input is less than or equal to 0.2.
